RIGIDITY OF SMOOTH PROPER FIBRES WITH DIAGONAL KERNEL RESTRICTION

THE STACKS PROJECT AUTHORS

Editorial note. Two forward references to same-chapter lemma-limit-arguments (judged citation-only this batch, not carded); transcribed as-is.

Allowed standard prerequisites (editorial).

- Orthogonal point sheaf. lemma-orthogonal-point-sheaf (carded this batch, L26488). Case I: kernel vanishes on an open neighbourhood off the diagonal.
- Koszul deformation lemma. lemma-deform-koszul (carded this batch, L26463). Case II: lifts the Koszul representation over the special fibre.
- Limit arguments glue. lemma-limit-arguments (same chapter, not carded). Spreads the local Koszul representation to an affine neighbourhood.
- Isomorphism in a neighbourhood. lemma-isomorphism-in-neighbourhood (carded this batch, L26474). Final step: fibre isomorphism spreads over the base.

Source reference. Original lemma 0G0N: sources/stacks-equiv/pinned-original-source.tex, source lines 3962–3962. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G0Q: sources/stacks-equiv/pinned-original-source.tex, source lines 4080–4080. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G0P: sources/stacks-equiv/pinned-original-source.tex, source lines 4039–4039. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G03: sources/stacks-equiv/pinned-original-source.tex, source lines 2795–2795. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 062F: focused-seeds/geometry/S9-eqv-no-deformations/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make

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is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0FDP: focused-seeds/geometry/S9-eqv-no-deformations/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let k be a field. Let S be a finite type scheme over k with k-rational point s. Let $Y \to S$ be a smooth proper morphism. Let $X = Y_s \times S \to S$ be the constant family with fibre Y_s. Let K be the Fourier-Mukai kernel of a relative equivalence from X to Y over S. Assume the restriction*

$$L(Y_s \times_S Y_s \to X \times_S Y)^* K \cong \Delta_{Y_s/k,*}\mathcal{O}_{Y_s}$$

in $D(\mathcal{O}_{Y_s \times Y_s})$. Then there is an open neighbourhood $s \in U \subset S$ such that $Y|_U$ is isomorphic to $Y_s \times U$ over U.

Proof. Denote $i : Y_s \times Y_s = X_s \times Y_s \to X \times_S Y$ the natural closed immersion. (We will write Y_s and not X_s for the fibre of X over s from now on.) Let $z \in Y_s \times Y_s = (X \times_S Y)_s \subset X \times_S Y$ be a closed point. As indicated we think of z both as a closed point of $Y_s \times Y_s$ as well as a closed point of $X \times_S Y$.

Case I: $z \notin \Delta_{Y_s/k}(Y_s)$. Denote $\mathcal{O}_z$ the coherent $\mathcal{O}_{Y_s \times Y_s}$-module supported at z whose value is $\kappa(z)$. Then $i_*\mathcal{O}_z$ is the coherent $\mathcal{O}_{X \times_S Y}$-module supported at z whose value is $\kappa(z)$. Our assumption means that

$$K \otimes^{\mathbf{L}}_{\mathcal{O}_{X \times_S Y}} i_*\mathcal{O}_z = Li^*K \otimes^{\mathbf{L}}_{\mathcal{O}_{Y_s \times Y_s}} \mathcal{O}_z = 0$$

Hence by Lemma 0G03 we find an open neighbourhood $U(z) \subset X \times_S Y$ of z such that $K|_{U(z)} = 0$. In this case we set $Z(z) = \emptyset$ as closed subscheme of $U(z)$.

Case II: $z \in \Delta_{Y_s/k}(Y_s)$. Since Y_s is smooth over k we know that $\Delta_{Y_s/k} : Y_s \to Y_s \times Y_s$ is a regular immersion, see More on Morphisms, Lemma 0FDP. Choose a regular sequence $\overline{f}_1, \ldots, \overline{f}_r \in \mathcal{O}_{Y_s \times Y_s, z}$ cutting out the ideal sheaf of $\Delta_{Y_s/k}(Y_s)$. Since a regular sequence is Koszul-regular (More on Algebra, Lemma 062F) our assumption means that

$$K_z \otimes^{\mathbf{L}}_{\mathcal{O}_{X \times_S Y, z}} \mathcal{O}_{Y_s \times Y_s, z} \in D(\mathcal{O}_{Y_s \times Y_s, z})$$

is represented by the Koszul complex on $\overline{f}_1, \ldots, \overline{f}_r$ over $\mathcal{O}_{Y_s \times Y_s, z}$. By Lemma 0G0N applied to $\mathcal{O}_{S,s} \to \mathcal{O}_{X \times_S Y, z}$ we conclude that $K_z \in D(\mathcal{O}_{X \times_S Y, z})$ is represented by the Koszul complex on a regular sequence $f_1, \ldots, f_r \in \mathcal{O}_{X \times_S Y, z}$ lifting the regular sequence $\overline{f}_1, \ldots, \overline{f}_r$ such that moreover $\mathcal{O}_{X \times_S Y}/(f_1, \ldots, f_r)$ is flat over $\mathcal{O}_{S,s}$. By some limit arguments (Lemma 0G0P) we conclude that there exists an affine open neighbourhood $U(z) \subset X \times_S Y$ of z and a closed subscheme $Z(z) \subset U(z)$ such that

(1) $Z(z) \to U(z)$ is a regular closed immersion,
(2) $K|_{U(z)}$ is quasi-isomorphic to $\mathcal{O}_{Z(z)}$,
(3) $Z(z) \to S$ is flat,
(4) $Z(z)_s = \Delta_{Y_s/k}(Y_s) \cap U(z)_s$ as closed subschemes of $U(z)_s$.

By property (2), for $z, z' \in Y_s \times Y_s$, we find that $Z(z) \cap U(z') = Z(z') \cap U(z)$ as closed subschemes. Hence we obtain an open neighbourhood

$$U = \bigcup\nolimits_{z \in Y_s \times Y_s \text{ closed}} U(z)$$

of $Y_s \times Y_s$ in $X \times_S Y$ and a closed subscheme $Z \subset U$ such that (1) $Z \to U$ is a regular closed immersion, (2) $Z \to S$ is flat, and (3) $Z_s = \Delta_{Y_s/k}(Y_s)$. Since $X \times_S Y \to S$ is proper, after replacing S by an open neighbourhood of s we may assume $U = X \times_S Y$. Since the projections $Z_s \to Y_s$ and $Z_s \to X_s$ are isomorphisms, we conclude that after shrinking S we may assume $Z \to Y$ and $Z \to X$ are isomorphisms, see Lemma 0G0Q. This finishes the proof. □